

Phase-2 Submission

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Github Repository

Link: https://github.com/Manjima-26/NM_MANJIMA_DS

1. Problem Statement

In an era dominated by digital streaming, users face an overwhelming number of movie choices. Navigating through this content without assistance often leads to choice fatigue and suboptimal viewing experiences. This project aims to solve this by creating an AI-powered movie recommendation system that uses collaborative and content-based filtering to provide personalized movie suggestions tailored to individual user preferences.

- **Problem Type:** Recommendation (ranking), with elements of regression (rating prediction) and clustering (user segmentation).
- **Impact:** Enhances user satisfaction, engagement, and retention on movie streaming platforms by helping users discover content they are most likely to enjoy.

2. Project Objectives

- To build a machine learning-based movie recommendation engine.
- To apply collaborative filtering and content-based filtering algorithms.
- To compare performance across models such as KNN, SVD, and Matrix Factorization.

- To improve accuracy through user and movie feature engineering.
- To produce explainable and real-time movie suggestions.

3. Flowchart of the Project Workflow



4. Data Description

- **Dataset:** MovieLens 100K
- **Source:** <https://grouplens.org/datasets/movielens/>
- **Type:** Structured (ratings.csv, movies.csv, users.csv)
- **Records:** 100,000 ratings from 943 users on 1682 movies
- **Target Variable:** rating (1–5 scale)
- **Static/Dynamic:** Static

5. Data Preprocessing

- Merged ratings, movies, and users data.

- *Removed duplicates and handled missing values.*
- *Converted genres into multi-hot encoded features.*
- *Encoded user IDs and movie IDs.*
- *Normalized rating scale if needed.*
- *Transformed data for input into matrix factorization models.*

6. Exploratory Data Analysis (EDA)

Univariate Analysis:

- *Distribution of ratings (most ratings are 4 and 5)*
- *Top genres and most-rated movies*

Bivariate/Multivariate:

- *Correlation heatmaps between user preferences and genres*
- *Average rating per user and per genre*

Insights:

- *Power users contribute majority of ratings.*
- *Romance and action are highly rated genres.*
- *Genre preferences vary significantly by user clusters.*

7. Feature Engineering

- *Created genre embeddings.*
- *Built user profile vectors based on past preferences.*
- *Applied PCA on user-genre matrix for dimensionality reduction.*
- *Engineered temporal features (e.g., rating time decay).*

8. Model Building

Models Used:

- ***KNN (Collaborative Filtering):*** Neighborhood-based recommendations
- ***SVD (Matrix Factorization):*** Latent factor model for sparse matrix completion
- ***Content-Based Filtering:*** Based on genres, tags, and user history

Training:

- *Split data 80/20*
- *Evaluated with RMSE and MAE for rating prediction*
- *Used precision@k and recall@k for top-N recommendation quality*

9. Visualization of Results & Model Insights

- **Confusion Matrix:** Not applicable for rating regression
- **RMSE Comparison Chart:** Visual comparison across models
- **Precision/Recall at K:** Bar plots showing model performance
- **Feature Importance:** Genre influence on user preferences

10. Tools and Technologies Used

- **Programming Language:** Python
- **IDE:** Jupyter Notebook
- **Libraries:** pandas, numpy, seaborn, matplotlib, scikit-learn, surprise, lightfm
- **Visualization Tools:** seaborn, matplotlib, Plotly

11. Team Members and Contributions

<i>S NO</i>	<i>TEAM MEMBERS</i>	<i>CONTRIBUTIONS</i>
<i>1.</i>	<i>Kaviya Bharathi K.</i>	<i>Data preprocessing</i>
<i>2.</i>	<i>Manjima M.</i>	<i>EDA, Evaluation</i>
<i>3.</i>	<i>Balaji G.</i>	<i>Feature engineering</i>
<i>4.</i>	<i>Vishnu Kumar K.</i>	<i>modeling</i>
<i>5.</i>	<i>Dhinesh Kumar B.</i>	<i>documentation</i>